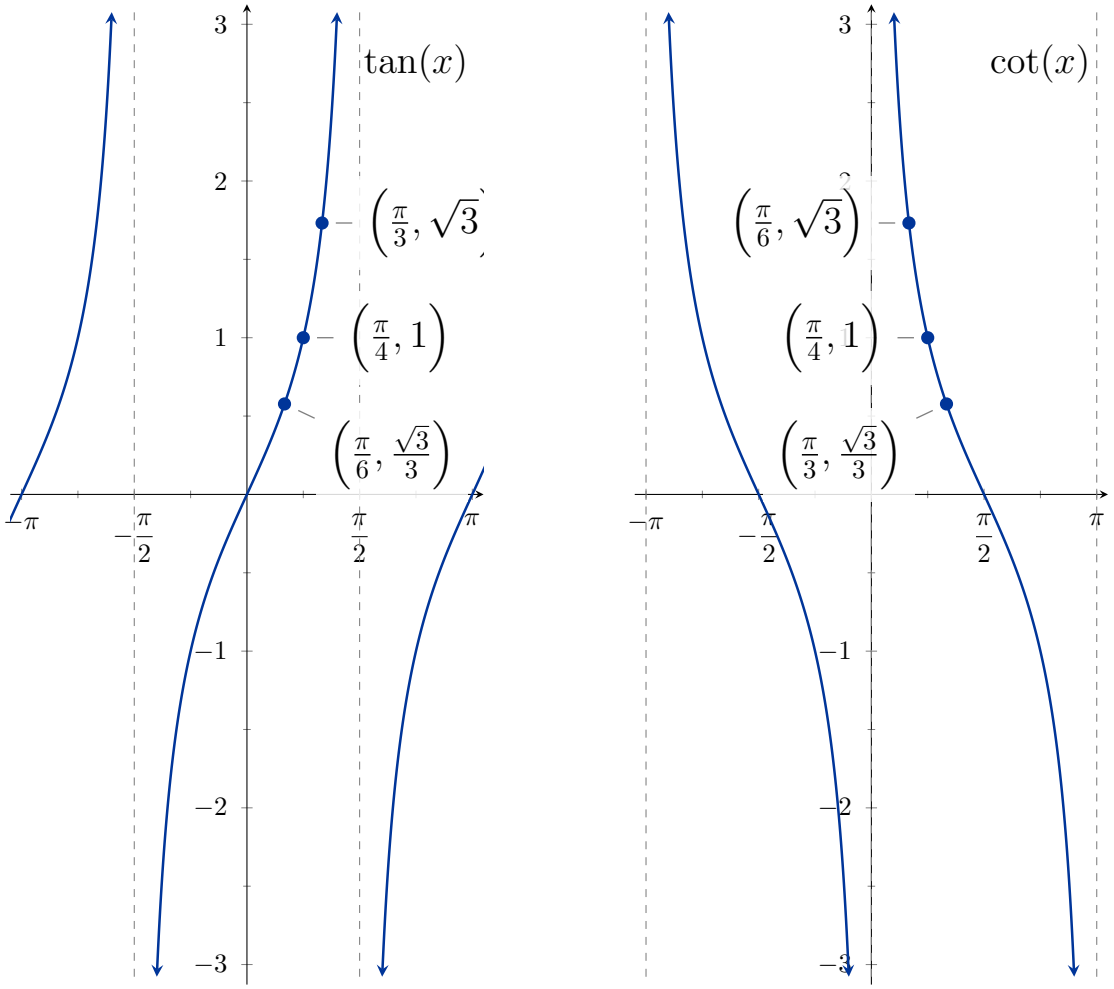
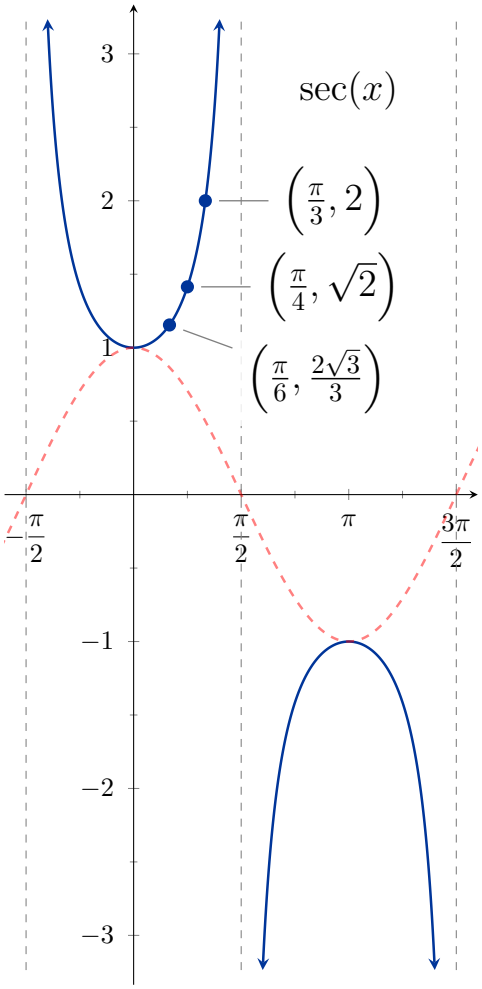
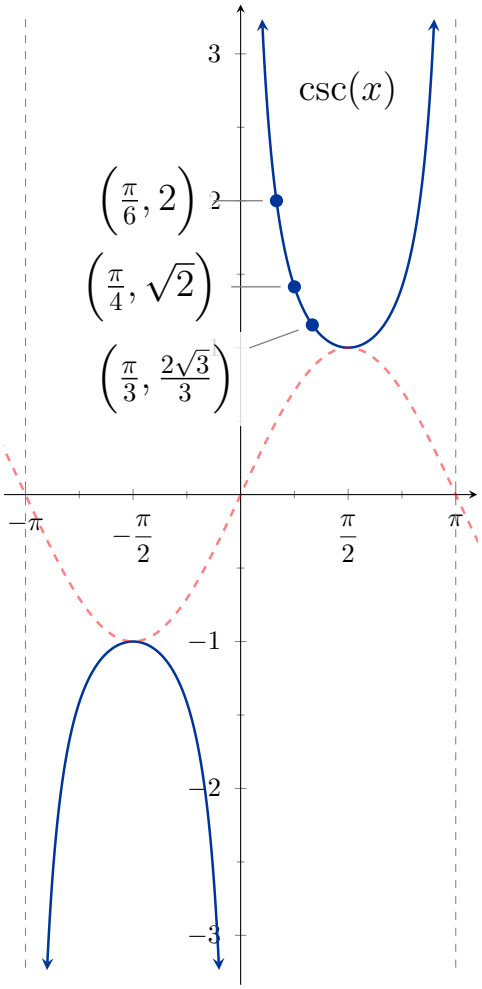
6.2: Graphs of the Other Trigonometric Functions




sec(x)
3
2
1
−1
−2
−3
−π/2
π/2
π
3π/2
(π/3, 2)
(π/4, √2)
(π/6, 2√3/3)


csc(x)
3
2
−1
−2
−3
−π
−π/2
π/2
π
(π/6, 2)
(π/4, √2)
(π/3, 2√3/3)

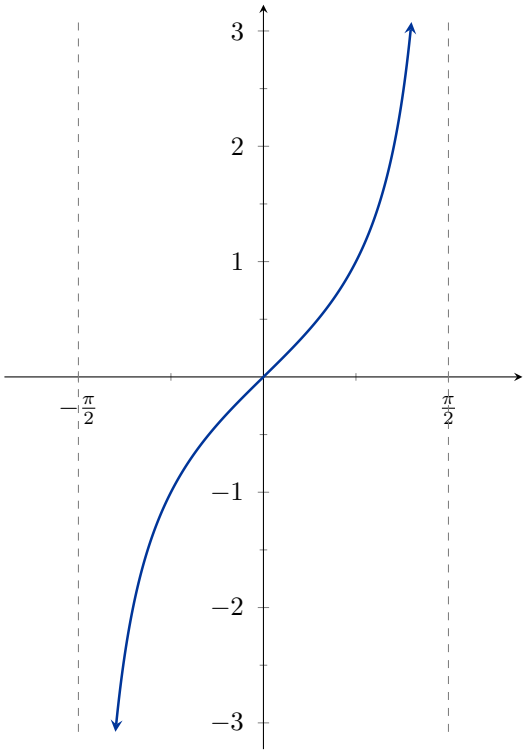
The general forms of the other trigonometric functions are

$$y = A\tan(Bx - C) + D, \qquad y = A\cot(Bx - C) + D,$$
$$y = A\sec(Bx - C) + D, \qquad y = A\csc(Bx - C) + D.$$

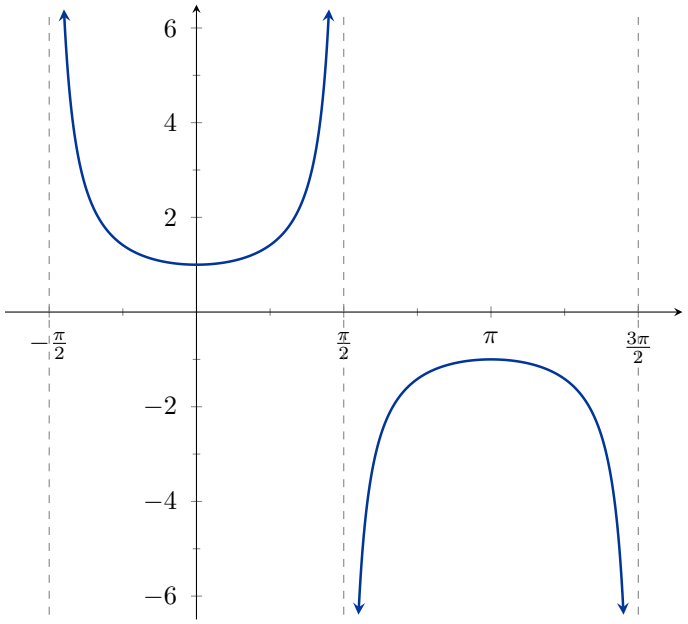
where

- A determines the scaling factor,
- B determines the period,
- C determines the phase shift, and
- D determines the vertical shift.

Example. Draw one period of the function $y = \frac{1}{2}\tan\left(\frac{\pi}{2}x\right)$



Example. Identify the domain, range, and draw one period of $y = 2.5 \sec\left(\frac{2}{5}x\right)$.



Example. Identify the domain, range, and draw one period of $y = -\cot\left(x + \frac{\pi}{2}\right)$.

